

Chapter - 1

Function

Definition of function:

Function is a relation from a set of inputs to a set of possible outputs where each input is related to exactly one output.

Algebraic function :-

classification of function:

1. Even Function $\rightarrow f(-x) = f(x)$

$$\text{ex: } y = \cos x + 13$$

2. Odd Function

$$\therefore y = \cos(-x) + 13$$

$$= \cos x + 13$$

$$\hookrightarrow f(x) = -f(x)$$

$$\text{ex: } y = x^3$$

$$\therefore y = (-x)^3$$

$$= -x^3$$

3. One-One Function (এক-এক প্রতিটি মানের জন্য)

4. Onto-Function : অবস্থানো মানের জন্য

Domain: The set of all possible input values, which produce a valid output from a particular function.

Range: The set of all possible values of the variable y , or sometimes expressed as $f(x)$, which result from using a particular function.

১) বৈশিষ্ট্য ২) Fractional ৩) Root. $x \in \mathbb{R}$

* Df and Rf

$y = f(x) = x + 13$

The function is defined for all real values of x . So, the domain of the given function is $Df = \mathbb{R}$.

$y = x + 13$

$x = y - 13$

x exists in its domain for all real values of y . So, the domain of the given function $Rf = \mathbb{R}$

Ans

1) $f(x) = x + |x - 1|$

$\therefore f(3) = 3 + |3 - 1|$

$= 3 + 2$

$= 5$

$f(-3) = -3 + |-3 - 1|$

$= -3 + |-4|$

$= -3 + 4$

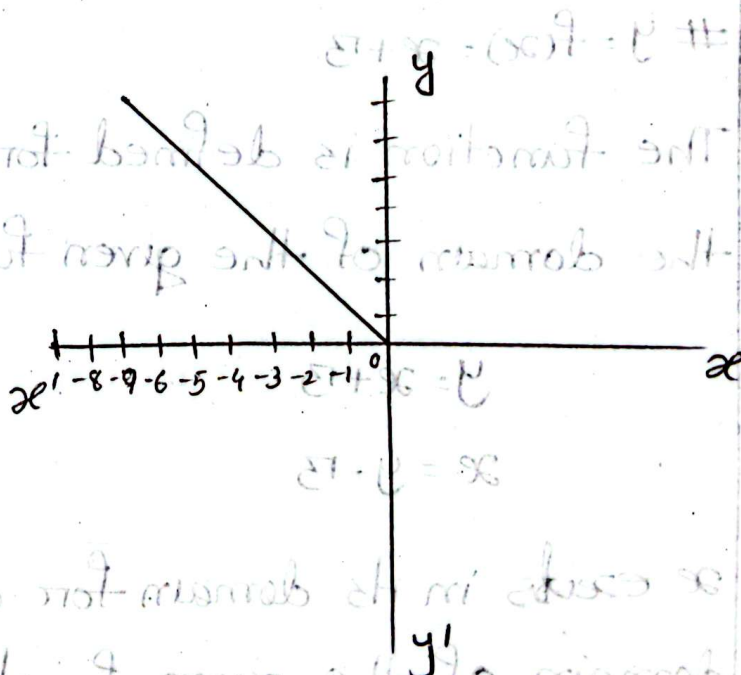
$= 1$

12) $f(x) = -x$ যাঃমানের লেখচিত্র অঙ্কন কর।

$f(x) = \begin{cases} -x & \text{when } x \leq 0 \\ x & \text{when } 0 < x < 1 \end{cases}$

x	-10	-9	-8	-5
$f(x)$	10	9	8	5

x	$1/4$	$1/3$	$1/2$	$3/4$
$f(x)$	$1/4$	$1/3$	$1/2$	$3/4$



25) let, $y = f(x) = \frac{x^2 - 4}{x - 2}$

The function is defined for all real values of x except

$$x = 2$$

So the domain of the given function is,

$$Df = \mathbb{R} - \{2\}$$

$$y = \frac{x^2 - 4}{x - 2} = x + 2$$

$$\therefore x = y - 2$$

$$\left. \begin{array}{l} 0 < x \text{ then } 1 - \\ 0 = x \text{ " } 0 \\ 0 < x \text{ " } 1 \end{array} \right\} = (x)$$

x exist is its domain for all real values of y except

except $y = 4$

So, the range of the given function is,

$$Rf = \mathbb{R} - \{4\}$$

28) (a) $f(x) = \sqrt{x-1}$

The function is defined for all $x \geq 1$.

So, the domain of the given function is, $Df = [1, \infty)$

$$y = -\sqrt{x-1}$$

$$\Rightarrow y^2 = x-1$$

$$\Rightarrow x = y^2 + 1$$

x exists in it's domain for all real values of y .
So, the range of the given function is $R_f = \mathbb{R}$.

Q6
$$f(x) = \begin{cases} -1 & \text{when } x < 0 \\ 0 & \text{u } x = 0 \\ 1 & \text{u } x > 0 \end{cases}$$

$$\begin{aligned} < = () \\ \leq = [] \\ = = \{ \} \end{aligned}$$

Ans:-

So, the domain of the given function is $D_f =$

$$= (-\infty, 0) \cup \{0\} \cup (0, \infty)$$

$$= (-\infty, \infty)$$

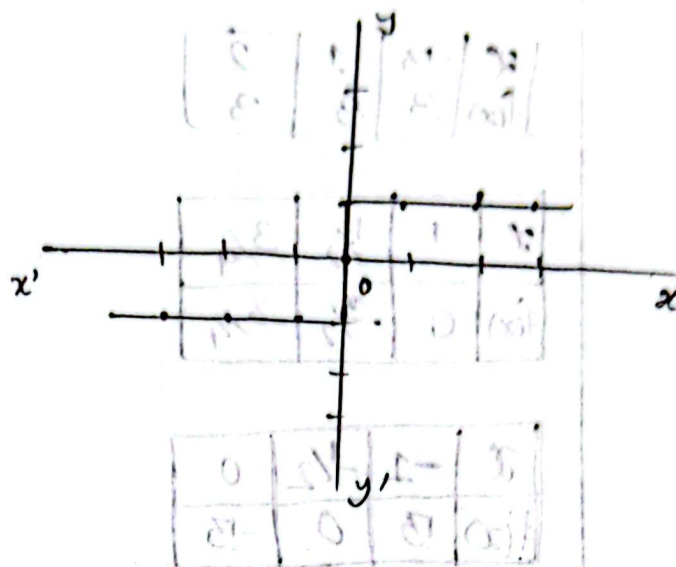
$$= \mathbb{R}$$

Thus the range of the given function is,

$$R_f = \{1, 0, -1\}$$

x	-3	-2	-1
$f(x)$	-1	-1	-1

x	1	2	3
$f(x)$	1	1	1



66) $f(x) = \begin{cases} 3 & \text{When } -3 \leq x < -1 \\ 3x-3 & \text{u } 0 < x \leq 1 \\ -6x-3 & \text{u } -1 \leq x \leq 0 \end{cases}$

Ans: Domain of the given function is, $D_f =$

$[-3, -1) \cup (0, 2] \cup [-1, 0]$

$= [-3, 1]$

For, $-3 \leq x < -1$, $f(x) = 3$

For, $0 < x \leq 1$, $-3 < f(x) \leq 0$

For, $-1 \leq x \leq 0$, $-3 \leq f(x) \leq 3$

Thus the range of the given function is $R_f =$

$\{3\} \cup (-3, 0] \cup [-3, 3] = [-3, 3]$

x	-3	-1	-2
$f(x)$	3	3	3

x	1	$\frac{1}{2}$	$\frac{3}{4}$
$f(x)$	0	$-\frac{3}{2}$	$-\frac{3}{4}$

x	-1	$-\frac{1}{2}$	0
$f(x)$	13	0	-13



80]

$$f(x) = \begin{cases} x^2 + 1, & \text{when } x < 0 \\ x, & \text{when } 0 \leq x \leq 1 \\ 1/x, & \text{when } 1 < x \end{cases}$$

Ans: $[0, 1] \cup [1, \infty) \cup (-\infty, 0]$

Domain of the given function is, $D_f =$

$$(-\infty, 0) \cup [0, 1] \cup (1, \infty)$$

$$= (-\infty, \infty)$$

$$= \mathbb{R}$$

For, $x < 0$, $f(x) > 1$.

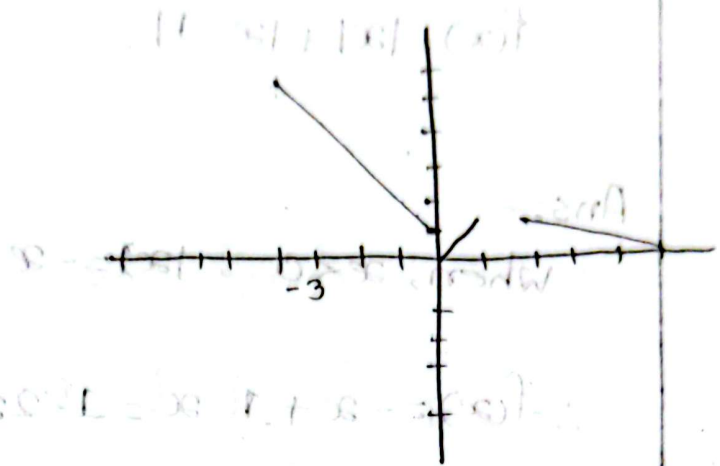
For, $0 \leq x \leq 1$, $0 \leq f(x) \leq 1$

For, $1 < x$, $0 < f(x) < 1$

x	-3	-2	-1	$-\frac{1}{2}$
$f(x)$	10	5	2	$\frac{5}{4}$

x	2	3	4	5
$f(x)$	$\frac{1}{2}$	$\frac{1}{3}$	$\frac{1}{4}$	$\frac{1}{5}$

x	0	$\frac{1}{2}$	1	
$f(x)$	0	$\frac{1}{2}$	1	



82] $f(x) = \begin{cases} x \sin(1/x), & \text{when } x \neq 0 \\ 0, & \text{when } x = 0 \end{cases}$

Ans: Domain of the given function is, $D_f = \mathbb{R} - \{0\} \cup \{0\} = \mathbb{R}$

Since, $x \neq 0$, the $\sin(1/x)$ oscillate between -1 to 1 .

Note: Sin, Cos

↓
Oscillating function
(अवमनमानक)

-tan

↓
range $[0, \infty)$

Range $[-1, 1]$

$$f(x) = |x| + |x-1|$$

Ans:-

When, $x < 0$, $|x| = -x$

$|x-1| = 1-x$

$$\therefore f(x) = -x + 1 - x = 1 - 2x$$

When, $0 \leq x < 1$, $|x| = x$ & $|x-1| = 1-x$

$$\therefore f(x) = x + 1 - x = 1$$

When, $x \geq 1$, $|x| = x$ & $|x-1| = x-1$

$$\therefore f(x) = x + x - 1 = 2x - 1$$

$$\therefore f(x) = \begin{cases} 1-2x & \text{when } x < 0 \\ 1 & \text{when } 0 \leq x < 1 \\ 2x-1 & \text{when } x \geq 1 \end{cases}$$

So, the domain of the given function is, $D_f =$

$$(-\infty, 0) \cup [0, 1) \cup [1, \infty) = (-\infty, \infty) = \mathbb{R}$$

for, $x < 0$, $f(x) > 1$ (increasing)

for, $0 \leq x < 1$, $f(x) = 1$

for, $x \geq 1$, $f(x) \geq 1$

Thus the range of the given function is, $\mathbb{R}^+$

$$(-\infty, 1) \cup \{1\} \cup [1, \infty) = [1, \infty)$$

Ans.

x	-1	-2	-3
$f(x)$	3	5	7

x	0	0.5	0.9
$f(x)$	1	1	1

x	1	2	3
$f(x)$	1	3	5

